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SpectroChangeNet Model for Change Detection in Synthetic Aperture Radar

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Abstract: Change detection using Synthetic Aperture Radar (SAR) has gained increased attention from the research community. The majority of existing change detection models have focused primarily on detection performance, with limited attention given to computational time and memory usage. To address this concern and achieve better change detection performance, a new deep learning model named SpectroChangeNet is introduced in this paper. The SpectroChangeNet model utilizes feature vectors from both the frequency and spatial domains to improve change detection results. Specifically, in the frequency domain, discrete coefficients are extracted from the SAR images using the Discrete Cosine Transform (DCT). In the spatial domain, the Multi-Region Convolution (MRC) module extracts deep feature vectors from the input images. The combined discrete coefficients and deep feature vectors are then used to classify uncertain pixels as changed or unchanged using the softmax function. In addition, a hybrid loss function is integrated with the MRC module to reduce the model's computational time and memory usage. The focal loss efficiently down-weights well-classified samples to concentrate more on hard or mis-classified uncertain samples. Further, the Mean Absolute Error (MAE) ensures that the gradient updates remain balanced and smooth. This model achieved 86.98%, 96.20%, and 94.28% of Kappa Coefficient (KC), and 96.36%, 98.92%, and 98.52% of Percentage of Correct Classification (PCC) on the Yellow River, Sulzberger, and Ottawa datasets. The achieved change detection results are superior compared to the existing models.

Keywords: Convolutional neural network, Change detection, Deep learning, Discrete cosine transform, Remote sensing, Synthetic aperture radar.

1. Introduction

Change detection is the process of analyzing and finding changes or differences in the earth's surface over time [1, 2]. In the present scenario, change detection has significant implications for various applications, such as disaster monitoring, land cover monitoring, object detection, deforestation monitoring, etc. [3-5]. Recently, several classical, unsupervised, and supervised change detection models have been developed for detecting changes in multi-temporal SAR images [6, 7]. Traditionally, classical methods like image rationing, image differencing, change vector analysis, etc. are used for SAR change detection. These classical change detection methods need accurate alignment between pre-event and post-event SAR images. Mis-registration due to temporal and geometric differences significantly reduces the performance of change detection [8, 9]. Furthermore, these classical change detection methods are ineffective at adapting to large-scale, heterogeneous, and complex datasets.

The supervised change detection methods, such as decision tree regression, random forest regression, etc., rely entirely on training samples to learn patterns of change [10, 11]. The performance of supervised change detection methods depends on preprocessing steps like feature normalization and speckle noise reduction [12, 13]. Numerous existing papers have shown that unsupervised change detection models are effective in capturing change information and have become a common solution for SAR change detection. However, in the application of change detection, the unsupervised models have the main concern of high computational time and memory usage, due to complex data processing requirements and model architecture. In this research paper, the computational bottleneck problem is highlighted by proposing a SpectroChangeNet model without compromising detection results. The major contributions of this paper are listed as follows;

- The speckle noise is efficiently suppressed by combining features from the spatial (MRC module) and frequency (DCT) domains. Specifically, in the MRC module, the contextual information and central region features are extracted from SAR images to achieve superior detection performance.
- A hybrid loss function is introduced in the SpectroChangeNet model to reduce overall computational time and memory usage by down-weighting well-classified samples and concentrating more on harder and mis-classified samples.
- The experiment conducted on three online datasets: Yellow River, Sulzberger, and Ottawa demonstrated the efficacy of the SpectroChangeNet model. The following five evaluation measures: KC, PCC, Overall Error (OE), memory usage, and computational time are used to validate the proposed model's performance.

This research paper is organized as follows: Section 2 illustrates the literature review of existing change detection models, Section 3 provides the theoretical and mathematical explanation of the SpectroChangeNet model, and Section 4 presents the conclusion of this paper.

2. Literature survey

Some of the existing literature is reviewed as follows: X. Qu, F. Gao, J. Dong, Q. Du, and H.C. Li [14] implemented a SAR change detection model called the Dual-Domain Network (DD-Net), which integrated features from both the spatial and frequency domains. The combination of DCT and MRC modules effectively alleviated speckle noise and achieved superior change detection results. Additionally, H. Zhang, Z. Lin, F. Gao, J. Dong, Q. Du, and H.C. Li [15] presented the Convolution and Attention Mixer (CAMixer) model for SAR change detection. The CAMixer model combined shift convolution and self-attention mechanisms to extract local features and global semantic information from SAR images, which helped achieve superior change detection results. D. Meng, F. Gao, J. Dong, Q. Du, and H.C. Li [16] developed a Layer Attention-based Noise Tolerant Network (LANTNet) model for change detection, in which a layer attention module was integrated into the traditional Convolutional Neural Network (CNN) to achieve superior detection performance. However, as depicted in the results section, the DD-Net, CAMixer, and LANTNet models consumed high computational time, particularly during the feature transformation process.

J. Xie, F. Gao, X. Zhou, and J. Dong [17] developed a novel SAR change detection model named the Wavelet-based Bi-dimensional Aggregation Network (WBA-Net). The WBA-Net model used Inverse Discrete Wavelet Transform (IDWT) and DWT for down-sampling without the loss of important image information. Furthermore, the bi-dimensional aggregation module effectively refined features and improved the capability of non-linear representation by capturing both channel and spatial dependencies. However, the dependence on complex aggregation processes and wavelet transforms increased computational time, which made this WBA-Net model less suitable for resource-constrained and real-time applications like change detection. L. Li, H. Ma, X. Zhang, X. Zhao, M. Lv, and Z. Jia [18] developed a framework for SAR change detection by combining the neighbor rule and fuzzy C-means clustering techniques with principal component analysis. This framework had a memory constraint problem when handling large datasets.

L. Ji, J. Zhao, and Z. Zhao [19] initially selected the training samples by performing an unsupervised pre-task. In the next step, the Siamese network was integrated with the superpixel clustering technique to obtain global optimal parameters. Finally, the obtained change map was superior compared to existing change detection models. This framework achieved better KC, overall accuracy, and F1-score by precisely refining change boundaries in small-area changes, but the developed framework was unsuitable for large datasets. X. Wang, X. Yan, K. Tan, C. Pan, J. Ding, Z. Liu, and X. Dong [20] implemented a double U-Net model comprising a superpixel module and a change detection module. The double U-Net model achieved superior change detection performance by maintaining the completeness of edge information and features. Although the integration of superpixel representation and change detection in the common network obtained a global optimal solution, it was still inappropriate for larger area changes due to high memory consumption and computational time. Additionally, L. Li, Q. Liu, G. Cao, L. Jiao, F. Liu, X. Liu, and P. Chen [21] introduced the Multidirectional Attention Fusion Network (MDAF-Net) model for

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SAR change detection. The integration of a multi-scale self-attention process with multi-directional feature learning significantly improved detection performance. However, the MDAF-Net was memory-intensive when dealing with higher-resolution SAR images.

To highlight the main concerns of computational time and memory usage with superior change detection performance on SAR images, a SpectroChangeNet model is proposed in this paper. This model employs a hybrid loss function, which integrates focal loss (focuses on hard-to-classify uncertain samples) and MAE (ensures smooth and balanced updates). This process effectively enhances detection accuracy, and reduces both computational time and memory usage. A detailed explanation of the SpectroChangeNet model is represented in the next section.

3. Methodology

As mentioned in the earlier sections, a novel deep learning model named SpectroChangeNet is proposed for SAR change detection. The primary objective of SpectroChangeNet model is to generate a change map (binary mask) by analyzing a pair of SAR images I_1 and I_2 acquired at varying times over the same geographic locale. In the resulting change map, unchanged pixels are represented as '0', while changed pixels are indicated as '1'. Initially, a high probability of unchanged and changed pixels is determined through pre-classification using the log-ratio operator. The log-ratio operator is applied to create a difference image, reducing the complexity of comparing the two SAR images feature-by-feature or pixel-by-pixel by isolating differences [22-23]. The log-ratio operator computes the ratio of pixel intensities between SAR images I_1 and I_2 in logarithmic terms, and it is mathematically denoted in Eq. (1) [24]. Where, $I_1(x,y)$ and $I_2(x,y)$ state the pixel intensities of the SAR images I_1 and I_2, log indicates the natural logarithm function, and $DI(x,y)$ represents the pixel value of the difference image.

$$DI(x,y) = log\left(\frac{I_1(x,y)}{I_2(x,y)}\right) \tag{1}$$

In the next step, a hierarchical Fuzzy C Means (FCM) clustering technique is employed for classifying the difference image into three clusters, namely Ω_c, Ω_u, and Ω_i [25-26]. Here, Ω_c and Ω_u state reliable pixels with a higher probability of being either changed or unchanged pixels. On the other hand, Ω_i comprises uncertain pixels that require further classification as either changed or unchanged pixels.

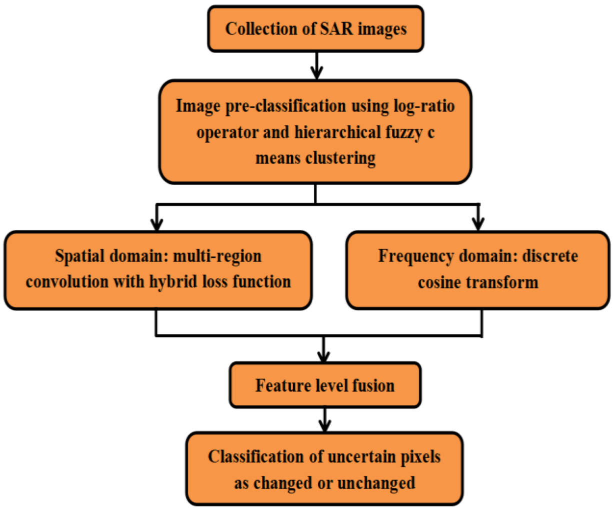

Figure. 1 Steps involved in the SpectroChangeNet model

After performing pre-classification, 10% of the image patches belonging to Ω_c and Ω_u are selected as training samples for the SpectroChangeNet model. The selected training samples should include an equal number of negative and positive samples, where the size of each image patch is $r \times r$ $(r = 7)$. By combining image patches from two SAR images I_1 and I_2, a new image patch is generated with a size of $2 \times r \times r$, which is fed to the SpectroChangeNet model for training. After training, the image patches centered on the pixels in Ω_i are classified by the proposed model as either changed or unchanged. The steps involved in the SpectroChangeNet model is presented in Fig. 1.

3.1 Spatial domain

In this domain, the SpectroChangeNet model comprises four MRC modules, which help to extract informative contextual information from the SAR images for accurate detection of changes. Generally, the existing unsupervised change detection models use fixed window sizes of $3 \times 3, 5 \times 5, 7 \times 7$, etc., for finding the status of the position, whether it is changed or unchanged.

The central region is emphasized, and the speckle noise in the marginal region is effectively reduced by abandoning a few marginal regions for feature extraction. Therefore, the MRC modules are introduced in the SpectroChangeNet model to extract contextual feature information from the SAR images for accurate change detection. Each MRC module includes three regions, namely the vertical middle

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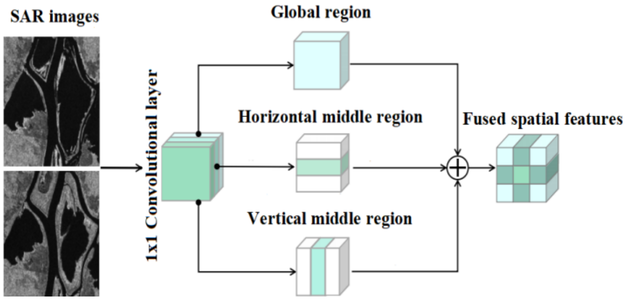

Figure. 2 Graphical presentation of single MRC module

region, horizontal middle region, and global region. Specifically, in the vertical middle region, the image pixels on the right and left are eliminated, and the image patches are designed to focus more on the center area. Similar to the vertical middle region, in the horizontal middle region, the image pixels on the bottom and top are eliminated, and here, the image patches are particularly designed to focus more on the center area. Lastly, the global region comprises square-shaped image patches, which guide the CNN in capturing the contextual feature information from the central pixel [27-28]. The graphical illustration of a MRC module is presented in Fig.2. An image patch size of $A \in \mathbb{R}^{2\times r\times r}$ is given as an input into the 1×1 convolutional layer for generating a feature map $F \in \mathbb{R}^{C\times r\times r}$. Furthermore, the generated feature map F is categorized into three groups, namely F_v, F_h, and F_g based on the channel dimension. In this scenario, the shape of F_v, F_h, and F_g is equal to $\frac{C}{3} \times r \times r$.

Here, F_v, F_h, and F_g are represented as the feature maps of the vertical middle region, horizontal middle region, and global region. In this research, the patch size r is fixed at 7, so two right columns and two left columns are fixed to 0 in F_v, and two bottom rows and two top rows are fixed to 0 in F_h. In the next step, three groups of features F'_v, F'_h, and F'_g are obtained after passing the feature maps of F_v, F_h, and F_g to the convolution layer 3×3. An element-wise summation is accomplished for fusing the obtained features, and this process is mathematically depicted in Eq. (2).

$$F_{fusion} = F'_v + F'_h + F'_g \tag{2}$$

Where, $F_{fusion} \in \mathbb{R}^{\frac{C}{3}\times r\times r}$ is represented as the combined spatial features. In this spatial domain, C is fixed at 15, so approximately $5 \times 7 \times 7 = 245$ features V_s are obtained from an image patch. In addition to this, a hybrid loss function (a combination of MAE and focal loss) is integrated with the MRC modules to ensure the gradient updates remain balanced and smooth, and to down-weight the well-classified uncertain samples Ω_i, focusing more on harder and mis-classified samples. This mechanism not only improves detection performance but also reduces computational time and memory usage. The hybrid loss function is mathematically expressed in Eq. (3). Here, τ_1 and τ_2 are hyper-parameters, which control the weighting of both MAE and focal loss.

$$Hybrid\_loss\_function = \tau_1 \times MAE + \tau_2 \times focal\_loss \tag{3}$$

3.2. Frequency domain

Generally, in the spatial domain, it is difficult to extract discriminative feature vectors from the difference image due to the presence of speckle noise. Therefore, in change detection applications, researchers focus more on the frequency domain, which effectively suppresses the speckle noise. In the frequency domain, DCT is employed to extract discriminative feature vectors from SAR images. First, a bi-linear interpolation is performed to resize the image patch from $2 \times r \times r$ to $2 \times 8 \times 8$. Next, the DCT is applied to convert the image patch into the frequency domain. Around 128 DCT coefficient vectors ϑ are obtained from the image patch, which is highly efficient in suppressing speckle noise. An on-off switch is used in the frequency domain to select the most crucial components from the DCT coefficient vectors ϑ. The on-off switch, along with two linear transformations and an attention gate g selects more informative feature vectors i for change detection. The mathematical expressions of the attention gate and the informative feature vectors are presented in Eq. (4) and (5).

$$g = \sigma(W^g\vartheta + b^g) \tag{4}$$

$$i = W^i\vartheta + b^i \tag{5}$$

Here, σ is represented as the sigmoid activation function, W^g and W^i are represented as the weight matrices, and b^i and b^g are stated as the bias values in the linear transformation. Using element-wise multiplication Θ, the DCT applies the attention gate to the informative feature vectors i for obtaining the final frequency based feature vectors V_f. Whereas, this process is mathematically illustrated in Eq. (6).

$$V_f = g \,\Theta\, i \tag{6}$$

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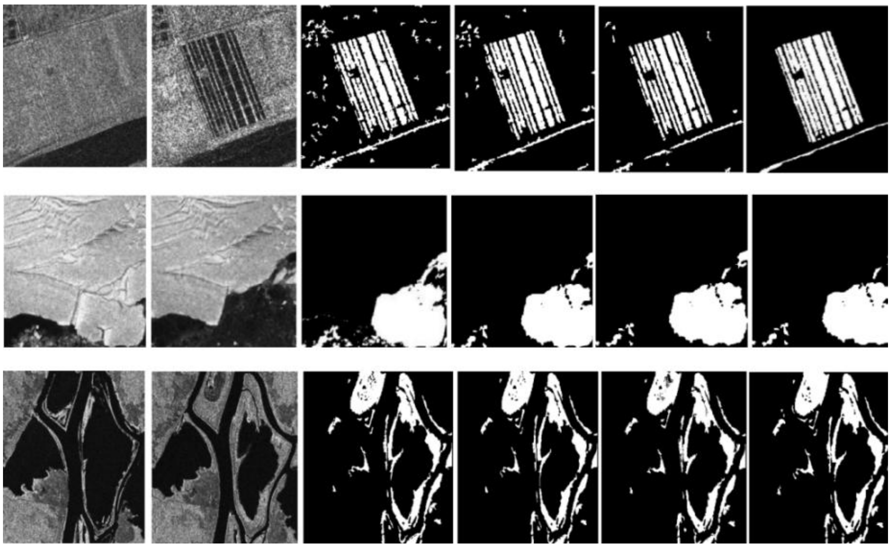

Figure 3. Yellow River dataset (1st row), Sulzberger dataset (2nd row), and Ottawa dataset (3rd row): (a) Input image I_1, (b) Input image I_2, (c) DD-Net, (d) LANTNet, (e) CAMixer, and (f) SpectroChangeNet

The obtained frequency domain feature vectors and spatial domain feature vectors are concatenated and further fed into the fully connected layer. Next, the possibility of unchanged and changed pixels is computed using the Softmax function for generating the output. After training the SpectroChangeNet model, the uncertain pixels Ω_i are classified as either unchanged or changed pixels, and lastly, the change map is obtained. The change detection results are visually represented in Fig. 3. The experimental results of the proposed SpectroChangeNet model are depicted in Section 4.

4. Results

The SpectroChangeNet model is implemented using the Python programming language on a system with 16GB of Random Access Memory (RAM), the Windows 11 operating system, a 1TB hard disk, and a GeForce RTX 3050 6GB Graphics Processing Unit (GPU). In the application of change detection, the performance of the SpectroChangeNet model is analyzed using five evaluation measures: KC, PCC, OE, computational time, and memory usage. The mathematical expressions of OE, PCC, and KC are represented in Eq. (7-10). Here, N represents the number of pixels in the SAR images I_1 and I_2. The terms True Negative (TN) and True Positive (TP) denote the number of unchanged and changed pixels correctly detected by the SpectroChangeNet model. Conversely, False Negative (FN) and False Positive (FP) represent the number of unchanged and changed pixels incorrectly detected by the proposed model. Further, NU and NC represent the number of unchanged and changed pixels in the change detection results.

$$OE = FP + FN \tag{7}$$

$$PCC = (TP + FN) / N \tag{8}$$

$$KC = PCC - Ƥ / 1 - Ƥ \tag{9}$$

Where,

$$Ƥ = (TP + FP) \times NC + (FN + TN) \times NU / N^2 \tag{10}$$

4.1 Dataset description

The effectiveness of the SpectroChangeNet model is tested on three online datasets: Yellow River, Sulzberger, and Ottawa [29-31]. Firstly, the Yellow River dataset was captured using the Radarsat-2 satellite between June 2008 and June 2009. The pixel size of the Yellow River dataset is 291×306. Secondly, the Sulzberger dataset was captured by the European space agency's envisat satellite, with a pixel size of 256×256. Finally, the Ottawa dataset was acquired using the Radarsat SAR satellite between May 1997 and August 1997, with a pixel size of 290×350.

4.2 Performance analysis

In this section, the SpectroChangeNet model is validated by varying patch sizes and loss functions.

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Table 1. Performance analysis of the SpectroChangeNet model with varying patch sizes

Patch size	Yellow River dataset			Sulzberger dataset			Ottawa dataset		
	OE	PCC (%)	KC (%)	OE	PCC (%)	KC (%)	OE	PCC (%)	KC (%)
5	2860	96.12	86.36	794	98.84	96.16	1768	98.44	94.10
7	2732	96.36	86.98	740	98.92	96.20	1520	98.52	94.28
9	2920	96.07	86.14	768	98.89	96.18	1790	98.40	94.05
11	3030	95.84	85.77	838	98.80	96.10	1850	98.36	93.89
13	3088	95.80	85.50	860	98.74	96.02	1904	98.28	93.82
15	3340	95.14	84.22	917	98.69	95.97	1976	98.12	93.76

Table 2. Performance analysis of the SpectroChangeNet model with varying loss functions

Loss function	Yellow River dataset			Sulzberger dataset			Ottawa dataset		
	OE	PCC (%)	KC (%)	OE	PCC (%)	KC (%)	OE	PCC (%)	KC (%)
MAE	2828	95.80	86.79	890	98.41	95.80	2028	97.40	93.75
Focal loss	2790	96.18	86.90	874	98.44	95.83	1962	97.66	93.82
Hybridization (MAE and focal loss)	2732	96.36	86.98	740	98.92	96.20	1520	98.52	94.28

Table 3. Comparative results of existing models and the SpectroChangeNet model by means of OE, PCC, and KC

Models	Yellow River dataset			Sulzberger dataset			Ottawa dataset		
	OE	PCC (%)	KC (%)	OE	PCC (%)	KC (%)	OE	PCC (%)	KC (%)
DD-Net [14]	2798	96.23	86.95	775	98.82	96.17	1668	98.36	93.77
CAMixer [15]	2764	96.28	86.86	-	-	-	-	-	-
LANTNet [16]	-	-	-	2013	96.93	91.87	1549	98.47	94.23
MDAF-Net [21]	2825	96.19	86.38	-	-	-	1684	98.34	93.72
SpectroChangeNet	2732	96.36	86.98	740	98.92	96.20	1520	98.52	94.28

First, the SpectroChangeNet model's performance is analyzed using different patch sizes: 5, 7, 9, 11, 13, and 15. By inspecting Table 1, it is evident that the SpectroChangeNet model achieved a lower overall error of 2732, a higher PCC of 96.36%, and a KC of 86.98% on the Yellow River dataset with a patch size of 7. These results indicate better change detection performance compared to other patch sizes. The OE, PCC, and KC values initially increase as the patch size increases, and then tend to stabilize. In most cases, some noise is introduced by using a large patch size. This process significantly affects the change detection results; therefore, a patch size of 7 is considered for implementation. Similar to the Yellow River dataset, the SpectroChangeNet model achieved a lower overall error of 740 and 1520, a higher PCC of 98.92% and 98.52%, and a KC of 96.20% and 94.28% on the Sulzberger and Ottawa datasets, respectively, with a patch size of 7. Further, in Table 2, the SpectroChangeNet performance is analyzed with varying loss functions: MAE, focal loss, and hybridization (MAE and focal loss). As stated in Table 2, the hybrid loss function achieves better change detection results compared to the individual loss functions. The use of the hybrid loss function in the SpectroChangeNet model dynamically reduced the importance of well-classified samples, making the model concentrate more on harder samples near the decision boundary. This process reduced the iterations required for fine-tuning the model, resulting in a slight reduction in computational time and memory usage. Additionally, this hybrid loss function ensures that gradient updates are more balanced by down-weighting easy samples, which leads to faster convergence. The combination of MAE and focal loss functions effectively stabilized the training process of the SpectroChangeNet model. This model achieved satisfactory change detection results with a minimal number of epochs.

4.3 Comparative analysis with discussion

As depicted in Table 3, four closely related change detection models (DD-Net, LANTNet, CAMixer, and MDAF-Net) are used for comparison to validate the efficacy of the SpectroChangeNet model. The existing change detection models, such as DD-Net, LANTNet, CAMixer, and MDAF-Net are implemented utilizing the default parameters provided in their respective works. The following parameters are fixed in the CAMixer model: patch

Table 4. Paired t-test analysis of PCC on the Ottawa, Sulzberger, and Yellow River datasets

Model comparison	Test statistic (t)	p-value	Interpretation
Ottawa dataset			
SpectroChangeNet vs DD-Net	2.78	0.045	Significant difference ($p < 0.05$)
SpectroChangeNet vs LANTNet	1.36	0.223	No significant difference
SpectroChangeNet vs MDAF-Net	3.12	0.033	Significant difference ($p < 0.05$)
Sulzberger dataset			
SpectroChangeNet vs DD-Net	2.45	0.049	Significant difference ($p < 0.05$)
SpectroChangeNet vs LANTNet	-1.12	0.298	No significant difference
Yellow River dataset			
SpectroChangeNet vs DD-Net	2.01	0.065	No significant difference
SpectroChangeNet vs CAMixer	2.14	0.053	Marginally significant difference
SpectroChangeNet vs MDAF-Net	2.58	0.036	Significant difference ($p < 0.05$)

Table 5. Comparative results of existing models and the SpectroChangeNet model in terms of computational time and memory usage

Models	Yellow River dataset		Sulzberger dataset		Ottawa dataset	
	Time (seconds)	Memory (MB)	Time (seconds)	Memory (MB)	Time (seconds)	Memory (MB)
DD-Net	26.12	18	24.09	17	30.11	22
CAMixer	25.44	22	23.58	21.40	29.38	24
LANTNet	25.10	17.80	23.40	17.20	29.20	31.30
MDAF-Net	28.22	22.40	26.40	22	34.28	33
SpectroChangeNet	24.38	12	22.20	11.30	28.50	14

size (9), loss function (cross entropy), optimizer (Adam), learning rate (3e-5), and total epochs (25). The parameters fixed in the DD-Net model are as follows: patch size (7), optimizer (Adam), loss function (cross entropy), total epochs (10), and learning rate (0.001). On the other hand, the parameters fixed in the LANTNet model are as follows: patch size (7), optimizer (Adam), loss function (cross entropy and MAE), total epochs (10), and learning rate (0.001). Furthermore, the MDAF-Net model includes following parameters: loss function (cross entropy), optimizer (Adam), and patch size (7). In this research paper, the parameters considered in the SpectroChangeNet model are as follows: patch size (7), optimizer (Adam), loss function (MAE and focal loss), total epochs (10), and learning rate (0.001). By inspecting Table 3, the SpectroChangeNet model achieved better change detection performance compared to existing models (DD-Net, LANTNet, CAMixer, and MDAF-Net) on three online datasets in terms of OE, PCC, and KC. Hence, the efficient design of the SpectroChangeNet model is adaptable to broader applications in the context of SAR change detection. It has a superior ability and plays a crucial role in detecting changes related to disaster management (landslides, earthquakes, and floods). Additionally, the SpectroChangeNet model is effective in monitoring land-use and land-cover changes, which provides better insights in agricultural activities, deforestation, and urbanization. Furthermore, the paired t-test analysis of PCC on the Ottawa, Sulzberger, and Yellow River datasets is given in Table 4. This table ensures that the SpectroChangeNet model is reliable in SAR change detection compared to other models.

As illustrated in Table 5, the SpectroChangeNet model consumed minimal computational time of 24.38, 22.20, and 28.50 seconds, and memory usage of 12 MB, 11.30 MB, and 14 MB on the Yellow River, Sulzberger, and Ottawa datasets. The results achieved in terms of computational time and memory usage are better compared to the existing change detection models: DD-Net, CAMixer, LANTNet, and MDAF-Net. These models consumed computational time of 26.12, 25.44, 25.10, and 28.22 seconds; 24.09, 23.58, 23.40, and 26.40 seconds; and 30.11, 29.38, 29.20, and 34.28 seconds on the Yellow River, Sulzberger, and Ottawa datasets. On the other hand, the memory usage of the DD-Net, CAMixer, LANTNet, and MDAF-Net models is 18 MB, 22 MB, 17.80 MB, and 22.40 MB; 17 MB, 21.40 MB, 17.20 MB, and 22 MB; and 22 MB, 24 MB, 31.30 MB, and 33 MB on the Yellow River, Sulzberger, and Ottawa datasets.

5. Conclusion

In this research paper, a deep learning model named SpectroChangeNet is proposed for SAR change detection. The SpectroChangeNet model

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integrates feature vectors from both the frequency and spatial domains to enhance the performance of change detection. In the frequency domain, a gating process is combined with the DCT to extract the frequency feature vectors. In the spatial domain, a MRC module is designed to extract the spatial feature vectors. Additionally, a hybrid loss function is implemented in the spatial domain that prioritizes harder samples near the decision boundary by down-weighting well-classified samples. This approach reduces fine-tuning iterations and enables faster convergence. The hybrid loss function effectively balances gradient updates and stabilizes model training, resulting in reduced computational time and memory usage. The extensive experimental investigation demonstrates that the SpectroChangeNet model achieves superior change detection results compared to existing models, namely DD-Net, LANTNet, CAMixer, and MDAF-Net. The SpectroChangeNet model obtains 96.36%, 98.92%, and 98.52% of PCC, and 86.98%, 96.20%, and 94.28% of KC on the Yellow River, Sulzberger, and Ottawa datasets with limited computational time and memory consumption. In future work, the proposed change detection model will be tested on real-time, large-scale datasets. Additionally, generative artificial intelligence models can be employed for enhancing the proposed model's change detection performance.

Conflicts of interest

The authors declare no conflict of interest.

Author contributions

The paper conceptualization, method, experimental investigation, writing original draft, visualization and editing have been done by 1st author. The supervision and project administration have been done by 2nd author.

References

[1] V.S. Vudattu, and U.C. Pati, "Change Detection in Synthetic Aperture Radar Images based on a Spatial Pyramid Pooling Attention Network (SPPANet)", *Remote Sensing Letters*, Vol.14, No.11, pp.1139-1149, 2023.

[2] C. Zhao, L. Ma, L. Wang, T. Ohtsuki, P.T. Mathiopoulos, and Y. Wang, "SAR image change detection in spatial-frequency domain based on attention mechanism and gated linear unit", *IEEE Geoscience and Remote Sensing Letters*, Vol.20, pp.1-5, 2023.

[3] Y. Feng, J. Jiang, H. Xu, and J. Zheng, "Change detection on remote sensing images using dual-branch multilevel intertemporal network", *IEEE Transactions on Geoscience and Remote Sensing*, Vol.61, pp.1-15, 2023.

[4] R. Zhang, H. Zhang, X. Ning, X. Huang, J. Wang, and W. Cui, "Global-aware siamese network for change detection on remote sensing images", *ISPRS journal of photogrammetry and remote sensing*, Vol.199, pp.61-72, 2023.

[5] L. Ding, K. Zhu, D. Peng, H. Tang, K. Yang, and L. Bruzzone, "Adapting segment anything model for change detection in VHR remote sensing images", *IEEE Transactions on Geoscience and Remote Sensing*, Vol.62, 2024.

[6] J. Zhang, Y. Liu, B. Wang, and C. Chen, "A hierarchical fusion SAR image change-detection method based on HF-CRF model", *Remote Sensing*, Vol.15, No.11, pp.2741, 2023.

[7] X. Hou, Y. Bai, Y. Xie, H. Ge, Y. Li, C. Shang, and Q. Shen, "Deep collaborative learning with class-rebalancing for semi-supervised change detection in SAR image", *Knowledge-Based Systems*, Vol.264, pp.110281, 2023.

[8] J. Wang, F. Zeng, A. Zhang, and T. You, "A global patch similarity-based graph for unsupervised SAR image change detection", *Remote Sensing Letters*, Vol.15, No.4, pp.353-362, 2024.

[9] F. Luo, T. Zhou, J. Liu, T. Guo, X. Gong, and J. Ren, "Multiscale diff-changed feature fusion network for hyperspectral image change detection", *IEEE Transactions on Geoscience and Remote Sensing*, Vol.61, pp.1-13, 2023.

[10] Y. Wu, Q. Xu, X. Zhu, T. Zhao, B. Wen, and J. Ma, "Synthetic aperture radar image change detection based on image difference denoising and fuzzy local information C-means clustering", *Journal of Applied Remote Sensing*, Vol.18, No.2, pp.024501-024501, 2024.

[11] M.J. Al-Dujaili, "An accurate algorithm for land surface changes detection based on deep learning and improved pixel clustering using SAR images", *Neural Computing and Applications*, Vol.36, No.10, pp.5545-5554, 2024.

[12] T. Gao, H. Li, M. Gong, M. Zhang, and W. Qiao, "Superpixel-based multiobjective change detection based on self-adaptive neighborhood-based binary differential evolution", *Expert Systems with Applications*, Vol.212, pp.118811, 2023.

[13] L. Li, H. Ma, and Z. Jia, "Gamma correction-based automatic unsupervised change detection in SAR images via FLICM model", *Journal of*

the Indian Society of Remote Sensing, Vol.51, No.5, pp.1077-1088, 2023.

[14] X. Qu, F. Gao, J. Dong, Q. Du, and H.C. Li, “Change detection in synthetic aperture radar images using a dual-domain network”, *IEEE Geoscience and Remote Sensing Letters*, Vo1.19, pp.1-5, 2021.

[15] H. Zhang, Z. Lin, F. Gao, J. Dong, Q. Du, and H.C. Li, “Convolution and attention mixer for synthetic aperture radar image change detection”, *IEEE Geoscience and Remote Sensing Letters*, Vol.20, 2023.

[16] D. Meng, F. Gao, J. Dong, Q. Du, and H.C. Li, “Synthetic aperture radar image change detection via layer attention-based noise-tolerant network”, *IEEE Geoscience and Remote Sensing Letters*, Vol.19, pp.1-5, 2022.

[17] J. Xie, F. Gao, X. Zhou, and J. Dong, “Wavelet-based Bi-dimensional aggregation network for SAR image change detection”, *IEEE Geoscience and Remote Sensing Letters*, Vol.21, 2024.

[18] L. Li, H. Ma, X. Zhang, X. Zhao, M. Lv, and Z. Jia, “Synthetic aperture radar image change detection based on principal component analysis and two-level clustering”, *Remote Sensing*, Vol.16, No.11, pp.1861, 2024.

[19] L. Ji, J. Zhao, and Z. Zhao, “A novel end-to-end unsupervised change detection method with self-adaptive superpixel segmentation for SAR images”, *Remote Sensing*, Vol.15, No.7, pp.1724, 2023.

[20] X. Wang, X. Yan, K. Tan, C. Pan, J. Ding, Z. Liu, and X. Dong, “Double U-Net (W-Net): A change detection network with two heads for remote sensing imagery”, *International Journal of Applied Earth Observation and Geoinformation*, Vol.122, pp.103456, 2023.

[21] L. Li, Q. Liu, G. Cao, L. Jiao, F. Liu, X. Liu, and P. Chen, “Multidirectional Attention Fusion Network for SAR Change Detection”, *Remote Sensing*, Vol.16, No.19, pp.3590, 2024.

[22] W. Li, P. Ma, H. Wang, and C. Fang, “SAR-TSCC: A novel approach for long time series SAR image change detection and pattern analysis”, *IEEE Transactions on Geoscience and Remote Sensing*, Vol.61, pp.1-16, 2023.

[23] W. Li, B. Pang, X. Xu, and B. Wei, “A SAR change detection method based on an iterative guided filter and the log mean ratio”, *Remote Sensing Letters*, Vol.13, No.7, pp.663-671, 2022.

[24] C. Ghosh, D. Majumdar, and B. Mondal, “Detection of changes in synthetic aperture radar images using Modified Gauss-Log ratio and Fuzzy Local Information C-Means clustering”, *Multimedia Tools and Applications*, Vol.82, No.27, pp.42661-42678, 2023.

[25] Y. Peng, B. Cui, H. Yin, Y. Zhang, and P. Du, “Automatic SAR change detection based on visual saliency and multi-hierarchical fuzzy clustering”, *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, Vol.15, pp.7755-7769, 2022.

[26] F. Guo, J. Zhu, L. Huang, H. Li, J. Deng, H. Jiang, and X. Hou, “Enhancing Spatial Debris Material Classifying through a Hierarchical Clustering-Fuzzy C-Means Integration Approach”, *Applied Sciences*, Vol.13, No.8, pp.4754, 2023.

[27] U. Thirugnanam, N. Joseph, U. Srikanth, and R. Anand, “Convolutional Neural Network with Coordinate Attention Module for the Detection of Skin Cancer”, *International Journal of Intelligent Engineering & Systems*, Vol.17, No.3, 2024.

[28] G. Bompem, and D. Pandluri, “Batch Normalization Based Convolutional Neural Network for Segmentation and Classification of Brain Tumor MRI Images”, *International Journal of Intelligent Engineering & Systems*, Vol.17, No.2, 2024.

[29] H. Dong, T. Wang, H. Song, Z. Liu, and D. Zhou, “A Change Detection Approach Based on Multi-temporal SAR Images for Railway Perimeter Intelligent Perception”, *IEEE Transactions on Industrial Cyber-Physical Systems*, Vol.2, pp.435-445, 2024.

[30] S. Fang, C. Qi, S. Yang, Z. Li, W. Wang, and Y. Wang, “Unsupervised SAR Change Detection Using Two-Stage Pseudo Labels Refining Framework”, *IEEE Geoscience and Remote Sensing Letters*, Vol.21, pp.1-5, 2024.

[31] M. Gong, T. Gao, M. Zhang, W. Li, Z. Wang, and D. Li, “An M-nary SAR image change detection based on GAN architecture search”, *IEEE Transactions on Geoscience and Remote Sensing*, Vol.61, 2023.

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Notation list

Notation	Parameters
I_1 and I_2	SAR image_1 and image_2
$I_1(x, y)$ and $I_2(x, y)$	Pixel intensities of the SAR images
$\log$	Natural logarithm function
$DI(x, y)$	Difference image
Ω_c and Ω_u	Changed and unchanged pixels
Ω_i	Uncertain pixels
r	Patch size
A	Single image patch
F	Feature map
C	Channel dimension
F_v, F_h, and F_g	Feature maps of the vertical middle region, horizontal middle region, and global region
F_{fusion}	Combined spatial features
V_s	Spatial-based feature vectors
τ_1 and τ_2	Hyper-parameters of MAE and focal loss
ϑ	DCT coefficient vectors
g	Attention gate
i	Informative features in frequency domain
σ	Sigmoid activation function
W^g and W^i	Weight matrices
b^i and b^g	Bias values in the linear transformation
Θ	Element-wise multiplication
V_f	Frequency-based feature vectors
N	Number of pixels in the SAR images
NU and NC	Number of unchanged and changed pixels